The Distillation Idea

Pre-Work Module 08

ECBS5200 — Practical Deep Learning Engineering

The setup

You have **two models**:

- A **teacher**: big, expensive, accurate (e.g., 395M parameters)
- A **student**: small, fast, cheap (e.g., 149M parameters)

The teacher is too slow or too expensive to deploy.

The student is deployable — but not as good.

Question: Can the student learn from the teacher, not just from the data?

Hard labels vs. soft labels

Hard label (from the dataset):

> "This complaint is about **billing dispute**."

billing dispute	payment issue	account mgmt	credit report	fraud
1.0	0.0	0.0	0.0	0.0

Soft label (from the teacher model):

> "This is *probably* billing dispute, but it's *related* to payment issue."

billing dispute	payment issue	account mgmt	credit report	fraud
0.70	0.20	0.05	0.03	0.02

The soft label carries **more information**.

Why soft labels help

The teacher's probabilities encode **relationships between classes**.

- "billing dispute" and "payment issue" get similar probabilities → **they're related**
- "billing dispute" and "fraud" get very different probabilities → **they're not related**

Hard labels throw this information away. Every wrong class is equally wrong (0.0).

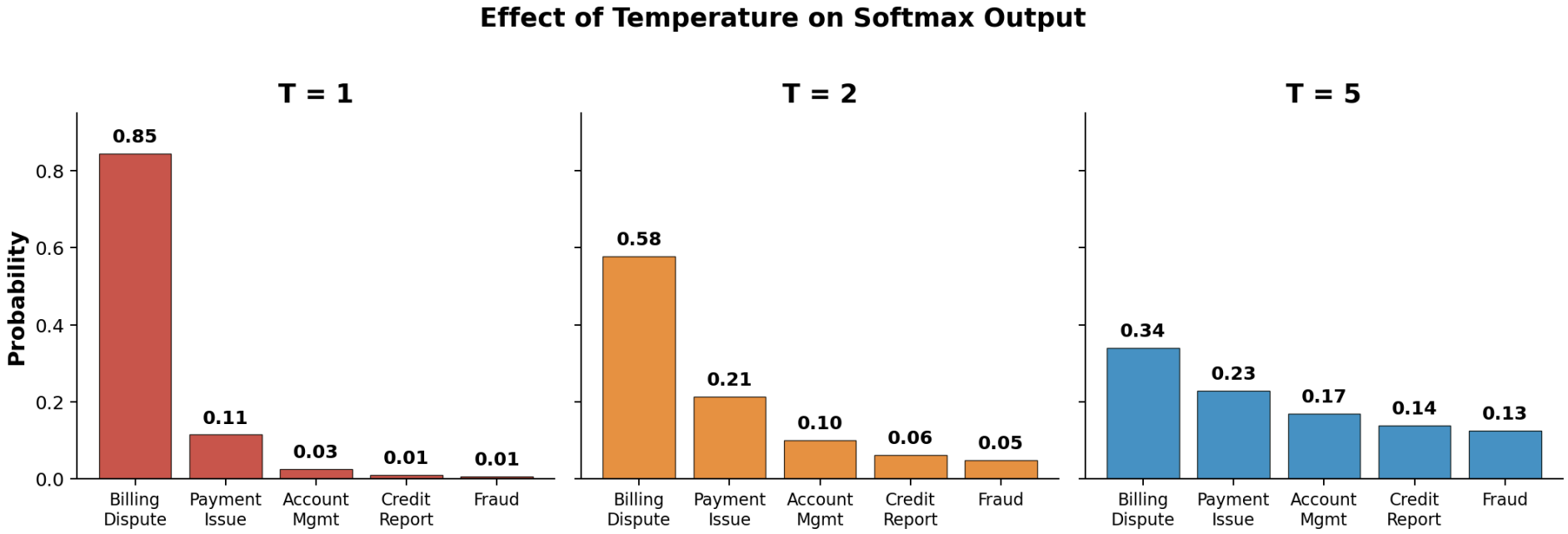
Soft labels preserve it. The student learns:

- **What the right answer is** (same as hard labels)
- **Which wrong answers are close** (bonus signal from soft labels)
- **The structure of the label space** (which classes cluster together)

This is called **dark knowledge** — information hidden in the teacher's mistakes.

Temperature: making soft labels softer

Softmax at **temperature T**: spread out the probabilities.



Higher temperature → **more uniform** → **more information about class relationships**

The course plan: Week 6

In Week 6, you will:

1. **Take a strong teacher model** (395M parameters) and run it on your training data
2. **Collect the teacher's soft predictions** — probability distributions for every example
3. **Train your student model** (149M, the model you've been building all semester) on a blend of:
    - **Hard labels**: the ground truth from the dataset
    - **Soft labels**: the teacher's probability distributions
4. **Compare** the distilled student to the student trained only on hard labels

The student inherits some of the teacher's knowledge — without getting bigger.

Key takeaways

1. **Distillation** = training a small model (student) using a big model's (teacher) predictions
2. **Soft labels** carry more information than hard labels — they encode class relationships
3. **Temperature** controls how spread out the soft labels are — higher T reveals more structure
4. **Dark knowledge** is the information hidden in the teacher's near-misses
5. **Week 6**: you'll distill a 395M teacher into your 149M student and measure the impact

The core insight: The teacher doesn't just know *what* the right answer is. It knows *how the wrong answers relate to each other*. That knowledge transfers.